



Session 11A: Trigonometric Foundations — Radians, the Unit Circle, and Six Functions

Phase 2 — Classical Techniques | 105 min

Covers: radians and degree conversion, unit circle coordinates, sin/cos/tan/csc/sec/cot definitions and graphs, inverse trig functions and their graphs

Prerequisite for: 11B — Trigonometric Identities, Equations, and Beyond

Part A: Radians and the Unit Circle — The Two Pillars

Example 1: What a Radian Really Is

Take a circle of radius r . Walk along the circumference for exactly r units of length. The angle you swept out is **1 radian**.

In one full trip around the circle, you walk $2\pi r$ (the full circumference). So there are 2π radians in a full circle.

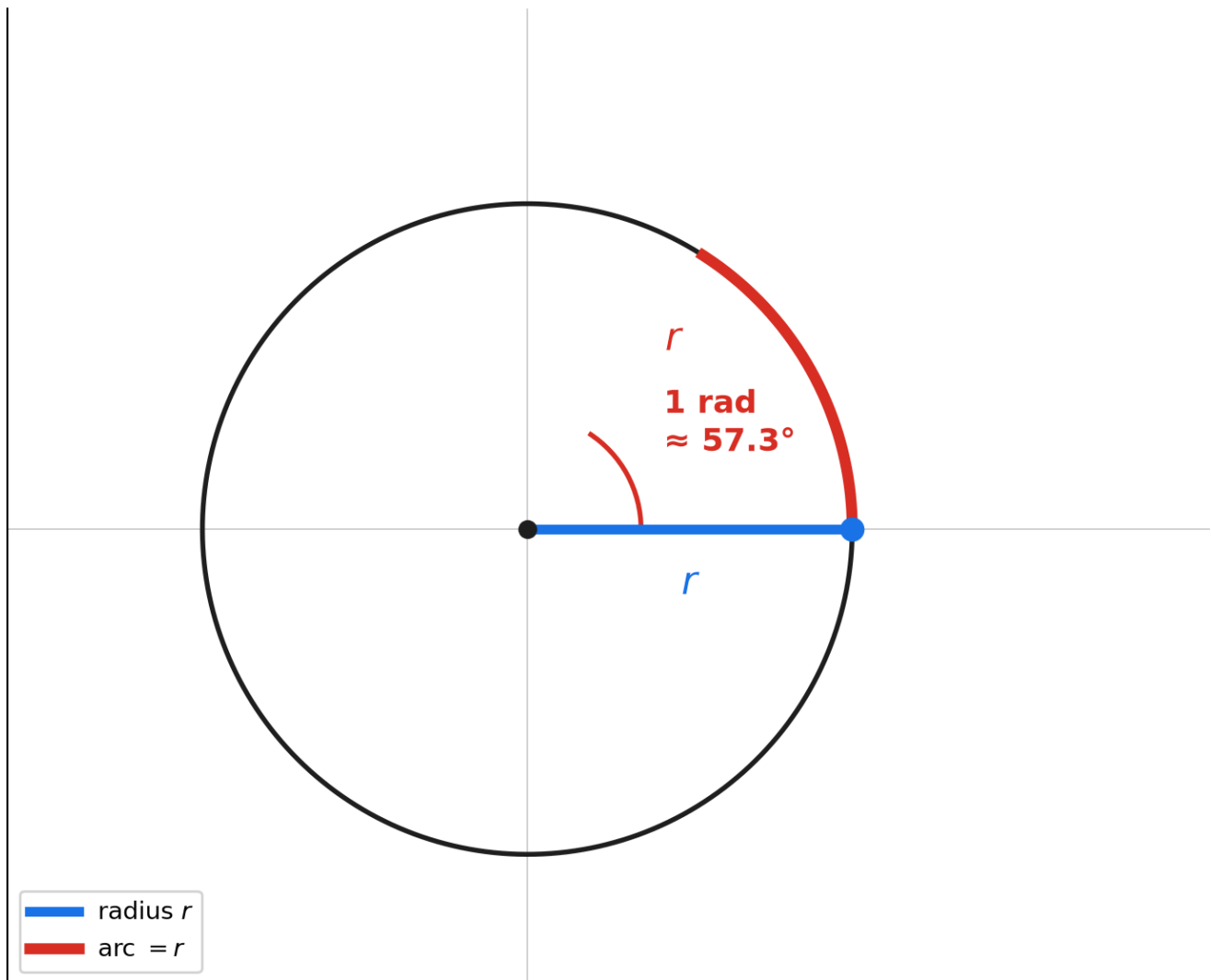
$$360^\circ = 2\pi \text{ rad. } 180^\circ = \pi \text{ rad. } 90^\circ = \frac{\pi}{2} \text{ rad.}$$

Why radians? Because the arc length formula becomes trivial: $s = r\theta$.

Degrees: $s = \frac{\pi r \theta}{180^\circ}$ — messy. Radians: $s = r\theta$ — clean.

This simplicity is why calculus *only* works in radians. $\frac{d}{dx} \sin x = \cos x$ is true only when x is in radians.

1 Radian — Arc Length = Radius



Geometric insight: 1 radian $\approx 57.3^\circ$. An angle of 1 radian cuts off an arc exactly as long as the radius. This is the natural unit of angle — no arbitrary "360" involved. The number 360 comes from ancient Babylonian astronomy (close to 365 days/year). Radians come from the geometry of the circle itself.

Example 2: Converting Between Degrees and Radians — Multiply by $\pi/180^\circ$

Degrees $\rightarrow$ radians: multiply by $\frac{\pi}{180^\circ}$.

$$30^\circ \rightarrow 30^\circ \times \frac{\pi}{180^\circ} = \frac{\pi}{6}.$$

$$45^\circ \rightarrow 45^\circ \times \frac{\pi}{180^\circ} = \frac{\pi}{4}.$$

$$60^\circ \rightarrow 60^\circ \times \frac{\pi}{180^\circ} = \frac{\pi}{3}.$$

$$120^\circ \rightarrow 120^\circ \times \frac{\pi}{180^\circ} = \frac{2\pi}{3}.$$

$$270^\circ \rightarrow 270^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{2}.$$

Radians $\rightarrow$ degrees: multiply by $\frac{180^\circ}{\pi}$.

$$\frac{\pi}{6} \rightarrow \frac{\pi}{6} \times \frac{180^\circ}{\pi} = 30^\circ.$$

$$\frac{5\pi}{4} \rightarrow \frac{5\pi}{4} \times \frac{180^\circ}{\pi} = 225^\circ.$$

$$\frac{7\pi}{3} \rightarrow \frac{7\pi}{3} \times \frac{180^\circ}{\pi} = 420^\circ \text{ (one full circle + } 60^\circ \text{)}.$$

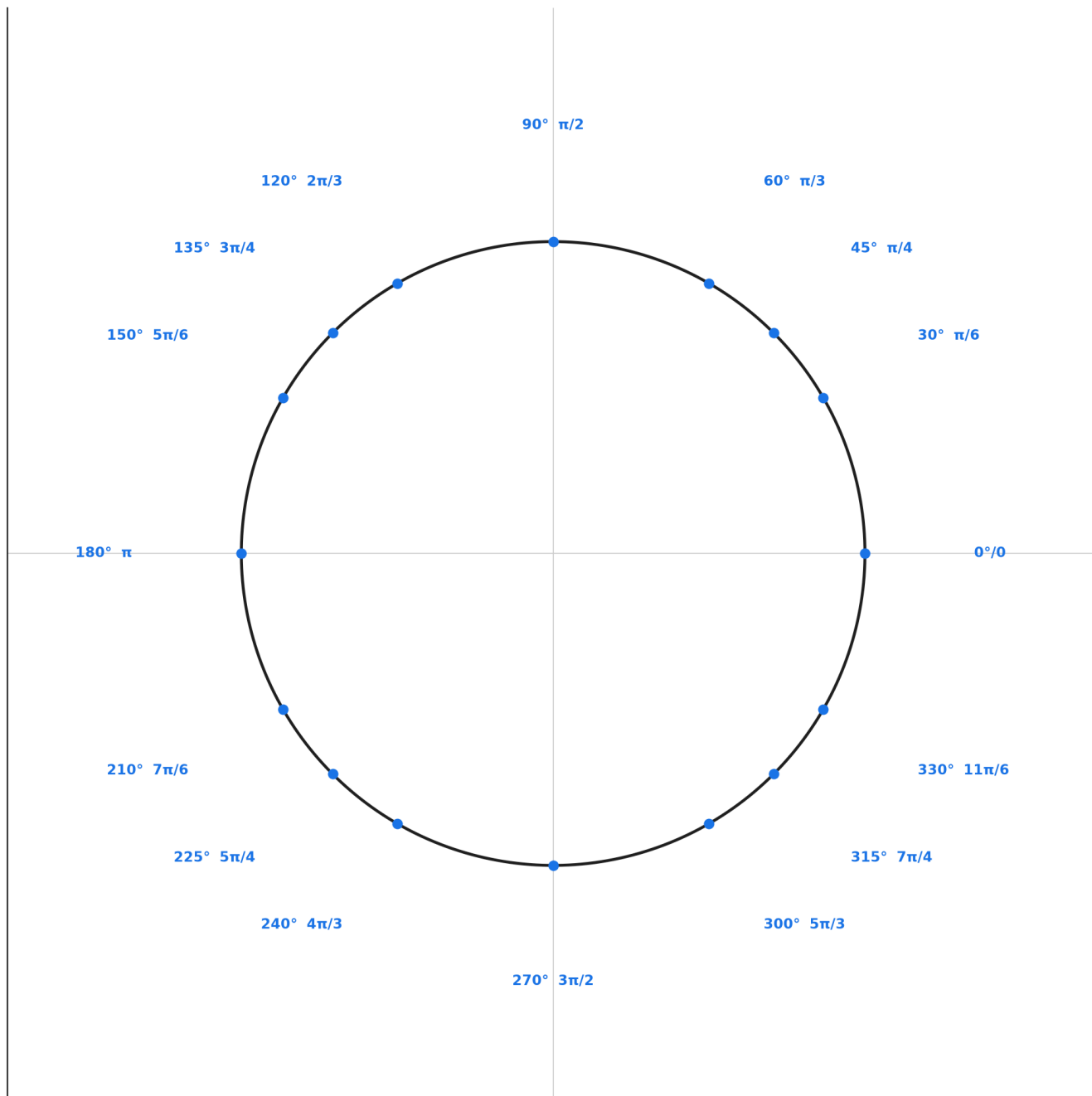
Mental shortcut table — memorize these and derive others by proportions:

Degrees	0°	30°	45°	60°	90°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

$$\text{To get } 120^\circ: 120^\circ = 2 \times 60^\circ = 2 \times \frac{\pi}{3} = \frac{2\pi}{3}.$$

$$\text{To get } 210^\circ: 210^\circ = 180^\circ + 30^\circ = \pi + \frac{\pi}{6} = \frac{7\pi}{6}.$$

Degrees ↔ Radians on the Unit Circle



Geometric insight: Thinking in radians means thinking in multiples of π . A quarter-turn is $\pi/2$, not 90. A half-turn is π , not 180. This rewires your brain to see angles as *distances along the circle*, not arbitrary numbers.

Method — Converting any angle in 3 steps:

(1) **Pick the direction.** Degrees $\rightarrow$ radians: multiply by $\frac{\pi}{180^\circ}$. Radians $\rightarrow$ degrees: multiply by $\frac{180^\circ}{\pi}$.

(2) **Multiply numerators, multiply denominators.** Treat π like a symbol, not a number.

$150^\circ \times \frac{\pi}{180^\circ} = \frac{150\pi}{180}$. Reduce the fraction: $\frac{150}{180} = \frac{5}{6}$. Result: $\frac{5\pi}{6}$.

(3) **If the result is $> 2\pi$ or negative, wrap it.** $\frac{7\pi}{3} > 2\pi$: subtract $2\pi = \frac{6\pi}{3} \rightarrow \frac{\pi}{3}$. $-\frac{\pi}{4}$ is negative: add $2\pi = \frac{8\pi}{4} \rightarrow \frac{7\pi}{4}$.

Example 3: The Unit Circle — A Circle of Radius 1

Draw a circle centered at $(0, 0)$ with radius 1. This is the **unit circle**.

Pick any angle θ measured counterclockwise from the positive x -axis. Draw a ray at that angle. It hits the circle at exactly one point. Call that point (x, y) .

The x -coordinate of that point is $\cos \theta$. The y -coordinate of that point is $\sin \theta$.

$\theta = 0$: ray hits $(1, 0)$. So $\cos 0 = 1$, $\sin 0 = 0$.

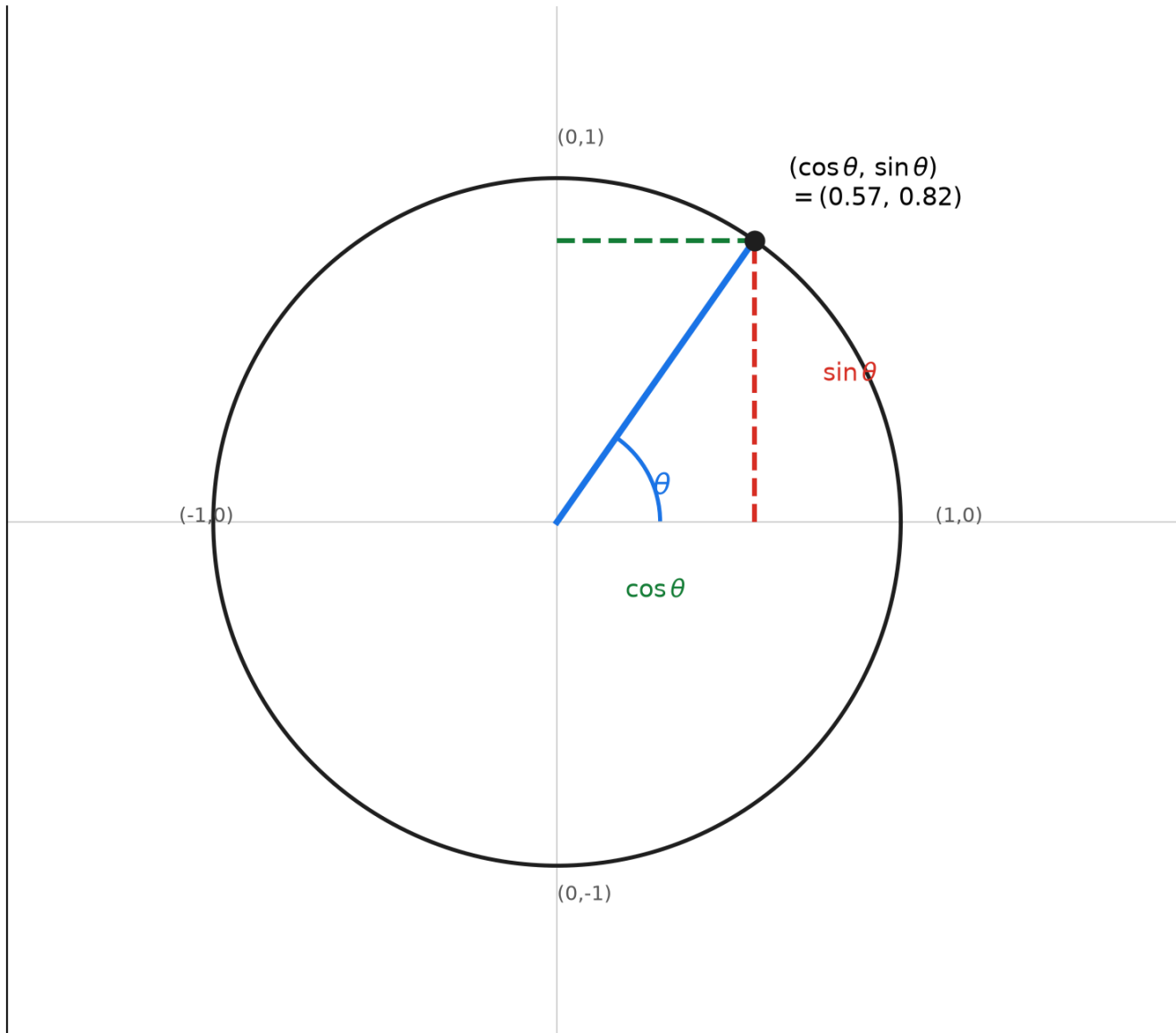
$\theta = \frac{\pi}{2}$: ray hits $(0, 1)$. So $\cos \frac{\pi}{2} = 0$, $\sin \frac{\pi}{2} = 1$.

$\theta = \pi$: ray hits $(-1, 0)$. So $\cos \pi = -1$, $\sin \pi = 0$.

$\theta = \frac{3\pi}{2}$: ray hits $(0, -1)$. So $\cos \frac{3\pi}{2} = 0$, $\sin \frac{3\pi}{2} = -1$.

The Pythagorean identity drops out immediately: $x^2 + y^2 = 1$ (every point on the unit circle satisfies this). Since $x = \cos \theta$ and $y = \sin \theta$: $\cos^2 \theta + \sin^2 \theta = 1$.

Unit Circle — $\cos \theta = x$, $\sin \theta = y$



Geometric insight: The unit circle turns trigonometry into coordinate geometry. $\cos \theta$ is just the horizontal position, $\sin \theta$ is the vertical position. Every trig fact about $\sin$ and $\cos$ is a fact about where a point lies on a circle.

Example 4: Special Angles — Exact Coordinates on the Unit Circle

$\theta = \frac{\pi}{4}$ (45°): The point lies on the line $y = x$. Since $x^2 + y^2 = 1$ and $x = y$:

$$2x^2 = 1 \rightarrow x = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}. \text{ So } \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}, \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}.$$

$\theta = \frac{\pi}{6}$ (**30°**): The 30-60-90 triangle has sides in ratio $1 : \sqrt{3} : 2$ (short leg : long leg : hypotenuse). On the unit circle, the hypotenuse is 1, so divide by 2:

Short leg (opposite 30°, which is $\sin$) = $\frac{1}{2}$. Long leg (adjacent to 30°, which is $\cos$) = $\frac{\sqrt{3}}{2}$.

$$\sin \frac{\pi}{6} = \frac{1}{2}, \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}.$$

$\theta = \frac{\pi}{3}$ (**60°**): Swap the legs from 30°.

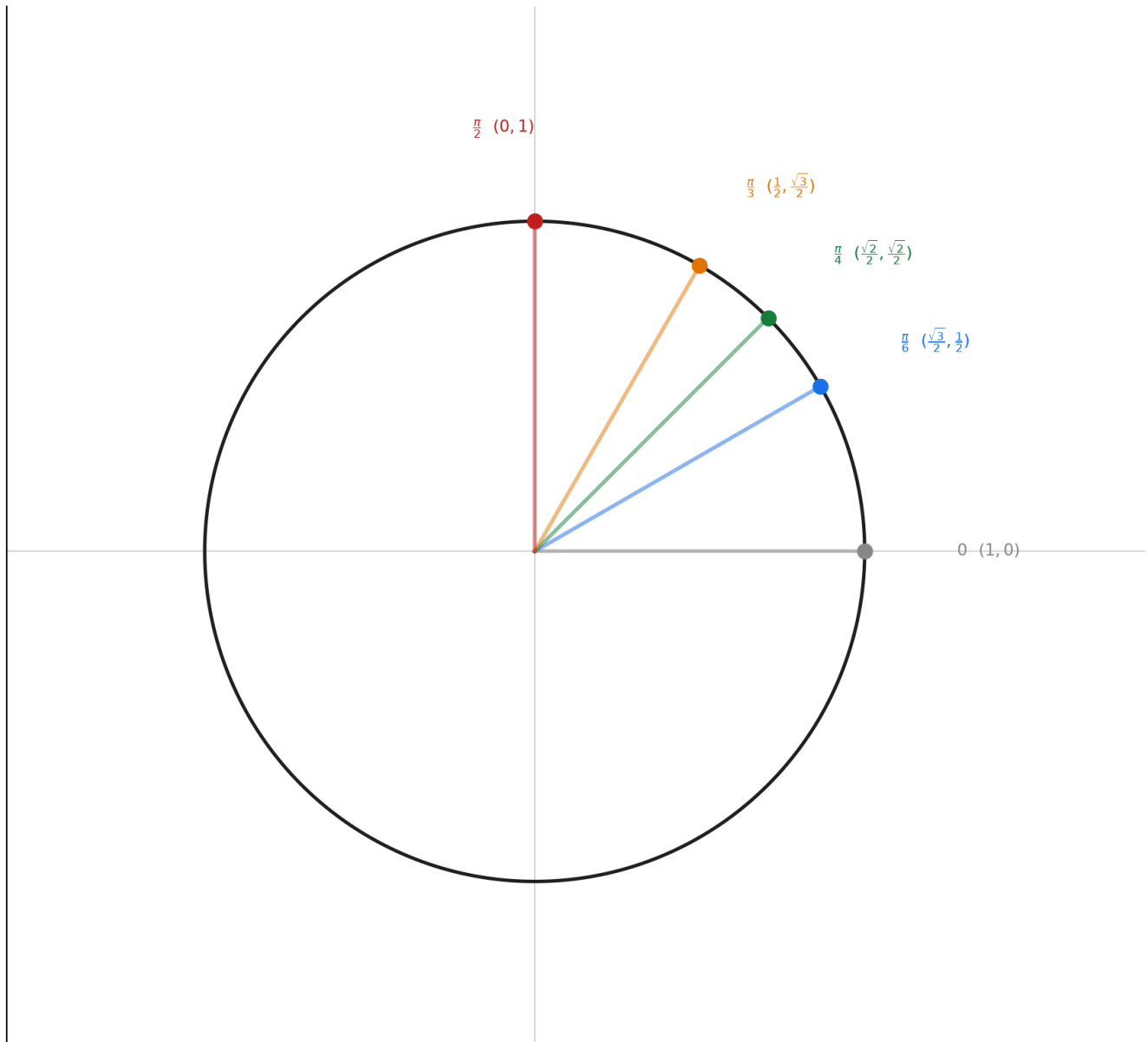
$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}, \cos \frac{\pi}{3} = \frac{1}{2}.$$

Summary table — first quadrant special angles:

θ (rad)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

Pattern to memorize: $\sin$ goes $0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$ (the $\frac{\sqrt{n}}{2}$ pattern for $n = 0, 1, 2, 3, 4$). $\cos$ is the same list read backwards.

Special Angles — Exact Coordinates



Geometric insight: Every angle that is a multiple of $\frac{\pi}{6}$ or $\frac{\pi}{4}$ has an exact coordinate expressible with square roots. These 16 points (4 quadrants $\times$ 4 angles each) are your reference grid. All other trig values flow from them via symmetry.

Example 5: Quadrants, Signs, and Reference Angles

Signs by quadrant — "All Students Take Calculus":

Quadrant	I	II	III	IV
sin	+	+	—	—
cos	+	—	—	+
tan	+	—	+	—

Memory: **A**ll positive in QI → **S**ine positive in QII → **T**angent positive in QIII → **C**osine positive in QIV.

Reference angle: For any angle θ , find the acute angle between the terminal ray and the x -axis. The $\sin/\cos$ *magnitudes* match the reference angle. The *sign* comes from the quadrant.

$$\theta = \frac{5\pi}{6} \text{ (150}^\circ\text{, QII)}. \text{ Reference angle} = \pi - \frac{5\pi}{6} = \frac{\pi}{6}.$$

$$\sin \frac{5\pi}{6} = + \sin \frac{\pi}{6} = \frac{1}{2} \text{ (positive in QII)}.$$

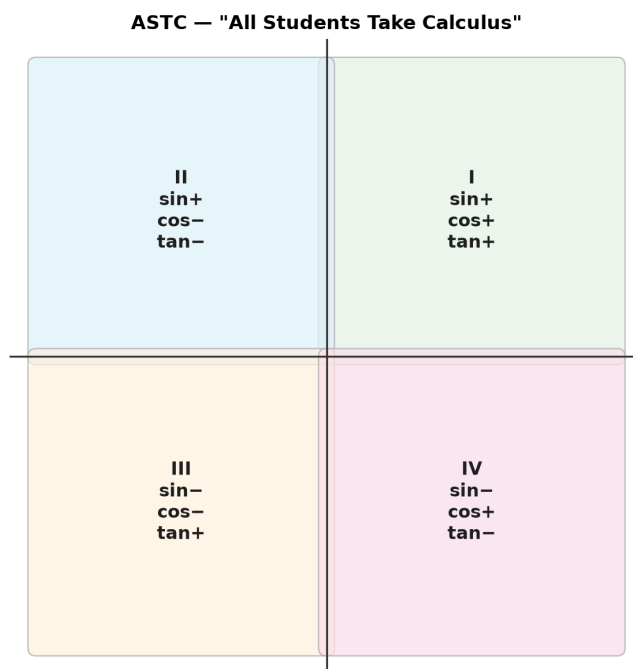
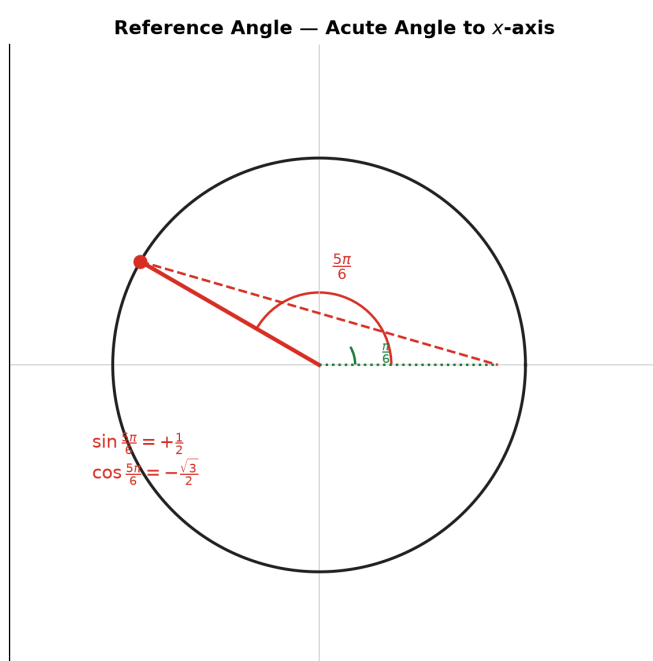
$$\cos \frac{5\pi}{6} = - \cos \frac{\pi}{6} = -\frac{\sqrt{3}}{2} \text{ (negative in QII)}.$$

$$\theta = \frac{7\pi}{4} \text{ (315}^\circ\text{, QIV)}. \text{ Reference angle} = 2\pi - \frac{7\pi}{4} = \frac{\pi}{4}.$$

$$\sin \frac{7\pi}{4} = -\frac{\sqrt{2}}{2}, \cos \frac{7\pi}{4} = +\frac{\sqrt{2}}{2}.$$

$$\theta = \frac{4\pi}{3} \text{ (240}^\circ\text{, QIII)}. \text{ Reference angle} = \frac{4\pi}{3} - \pi = \frac{\pi}{3}.$$

$$\sin \frac{4\pi}{3} = -\frac{\sqrt{3}}{2}, \cos \frac{4\pi}{3} = -\frac{1}{2}.$$



Geometric insight: Every angle outside QI is just a QI angle reflected across an axis. The reflection determines the sign. The reference angle determines the magnitude.

Up to here: Radians measure angles by arc length. Unit circle gives $\cos$ as x , $\sin$ as y . Special angles at multiples of $\frac{\pi}{6}$ and $\frac{\pi}{4}$ have exact radical coordinates. Reference angles + ASTC give any angle's trig values.

Method — Evaluating $\sin$, $\cos$, $\tan$ for any angle in 3 steps:

(1) **Normalize the angle into $[0, 2\pi)$.** If $\theta > 2\pi$, subtract 2π repeatedly until it falls in range. If $\theta < 0$, add 2π repeatedly.

(2) **Find the reference angle α .** Which quadrant? QI: $\alpha = \theta$. QII: $\alpha = \pi - \theta$. QIII: $\alpha = \theta - \pi$. QIV: $\alpha = 2\pi - \theta$. The reference angle is always acute (0 to $\frac{\pi}{2}$).

(3) **Read $\sin \alpha$, $\cos \alpha$ from the special-angle table. Attach the ASTC sign.**

QI: $(+, +)$. QII: $(+, -)$. QIII: $(-, -)$. QIV: $(-, +)$. Then $\tan = \sin / \cos$.

Walkthrough for $\frac{11\pi}{6}$: QIV. $\alpha = 2\pi - \frac{11\pi}{6} = \frac{\pi}{6}$. $\sin \frac{\pi}{6} = \frac{1}{2}$, $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$. QIV: $\sin (-)$, $\cos (+)$. $\rightarrow \sin \frac{11\pi}{6} = -\frac{1}{2}$, $\cos \frac{11\pi}{6} = \frac{\sqrt{3}}{2}$, $\tan \frac{11\pi}{6} = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$.

Part B: The Six Trigonometric Functions — Definitions and Graphs

Example 6: $\sin \theta$ and $\cos \theta$ — The Graphs Born from the Circle

Unroll the unit circle. As θ increases, the y -coordinate ($\sin \theta$) rises and falls, tracing a smooth wave. The x -coordinate ($\cos \theta$) does the same, shifted by $\frac{\pi}{2}$.

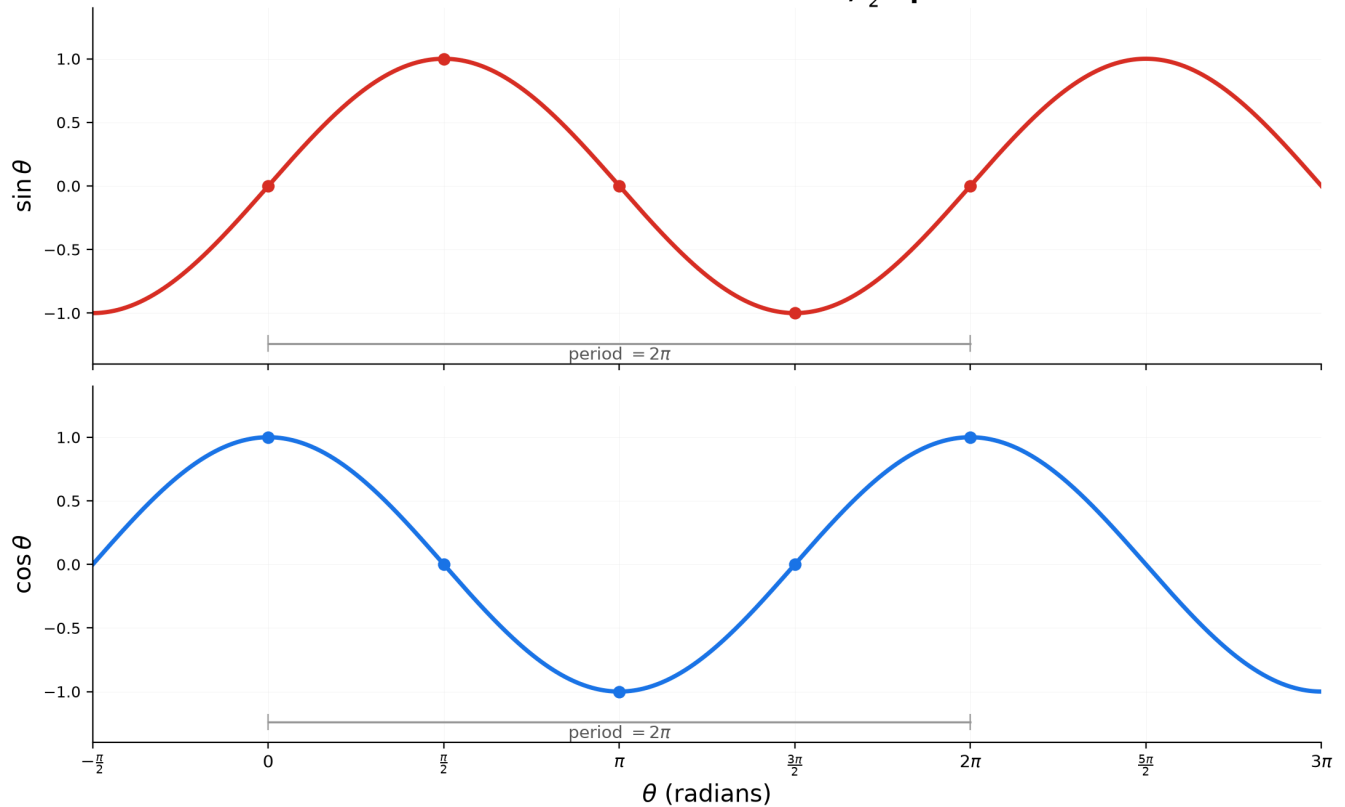
$\sin \theta$ graph:

- Starts at 0. Rises to 1 at $\frac{\pi}{2}$. Falls to 0 at π . Drops to -1 at $\frac{3\pi}{2}$. Returns to 0 at 2π .
- **Period** = 2π (one full wave repeats every 2π).
- **Amplitude** = 1 (the wave goes from -1 to 1).
- **Domain**: all real numbers. **Range**: $[-1, 1]$.
- **Symmetry**: $\sin(-\theta) = -\sin \theta$ (odd — symmetric about origin).

$\cos \theta$ graph:

- Starts at 1. Falls to 0 at $\frac{\pi}{2}$. Drops to -1 at π . Rises to 0 at $\frac{3\pi}{2}$. Returns to 1 at 2π .
- Same period and amplitude as sine.
- $\cos \theta = \sin(\theta + \frac{\pi}{2})$ — the cosine wave is the sine wave shifted left by $\frac{\pi}{2}$.
- **Symmetry**: $\cos(-\theta) = \cos \theta$ (even — symmetric about y -axis).

$\sin \theta$ and $\cos \theta$ — Period 2π , $\frac{\pi}{2}$ Apart



Geometric insight: Watch a point move around the unit circle at constant speed. Plot its y -coordinate against time — that's $\sin t$. Plot its x -coordinate against time — that's $\cos t$. The two waves are the same shape, just $\frac{\pi}{2}$ out of phase.

Example 7: $\tan \theta$ — Slope of the Ray

On the unit circle, $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{y}{x}$. This is the slope of the ray from the origin.

Geometrically, extend the ray until it hits the vertical line $x = 1$ (the tangent line to the circle at $(1, 0)$). The y -coordinate of that intersection is $\tan \theta$.

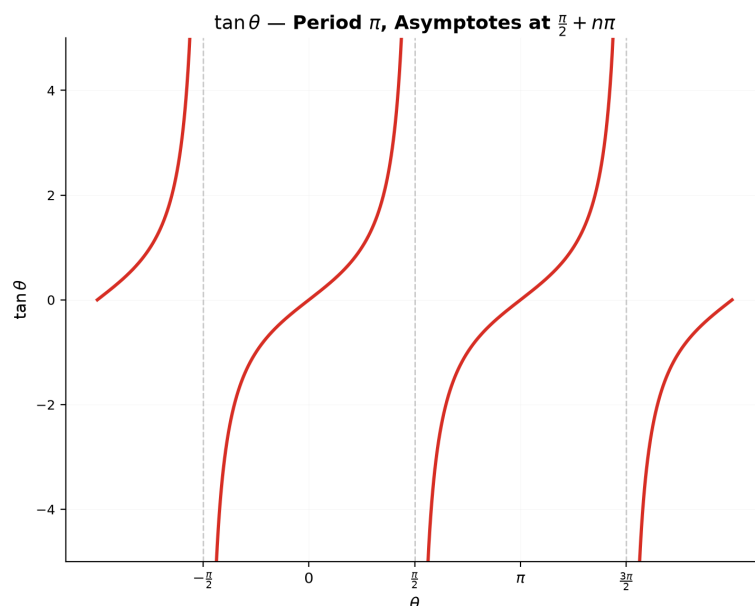
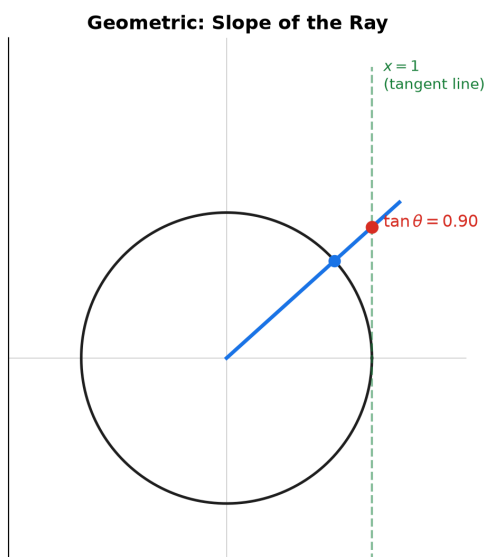
Special values:

$$\tan 0 = 0. \quad \tan \frac{\pi}{6} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$

$$\tan \frac{\pi}{4} = \frac{\sqrt{2}/2}{\sqrt{2}/2} = 1. \quad \tan \frac{\pi}{3} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}.$$

$\tan \theta$ graph:

- **Period** $= \pi$ (repeats every π , not 2π).
- **Vertical asymptotes** at $\theta = \frac{\pi}{2} + n\pi$ (where $\cos \theta = 0$).
- The graph shoots up to $+\infty$ approaching an asymptote from the left, and down from $-\infty$ from the right.
- **Domain**: all real numbers except $\frac{\pi}{2} + n\pi$. **Range**: all real numbers.
- **Symmetry**: $\tan(-\theta) = -\tan \theta$ (odd).



Geometric insight: $\tan \theta$ is the slope. As the ray gets nearly vertical, the slope explodes to infinity — hence the asymptotes. Every π radians you're back to the same slope (a line and its extension 180° later have the same slope), so the period is π .

Example 8: $\csc \theta$, $\sec \theta$, $\cot \theta$ — The Reciprocals

$$\csc \theta = \frac{1}{\sin \theta}, \sec \theta = \frac{1}{\cos \theta}, \cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}.$$

$\csc \theta$ graph (reciprocal of sine):

- Where $\sin \theta$ is large (± 1), $\csc \theta$ is small (± 1).
- Where $\sin \theta$ is near 0, $\csc \theta$ explodes to $\pm\infty \rightarrow$ vertical asymptotes at $\theta = n\pi$.
- U-shaped branches alternating above $y = 1$ and below $y = -1$.
- **Period** $= 2\pi$. **Range**: $(-\infty, -1] \cup [1, \infty)$.

$\sec \theta$ **graph** (reciprocal of cosine):

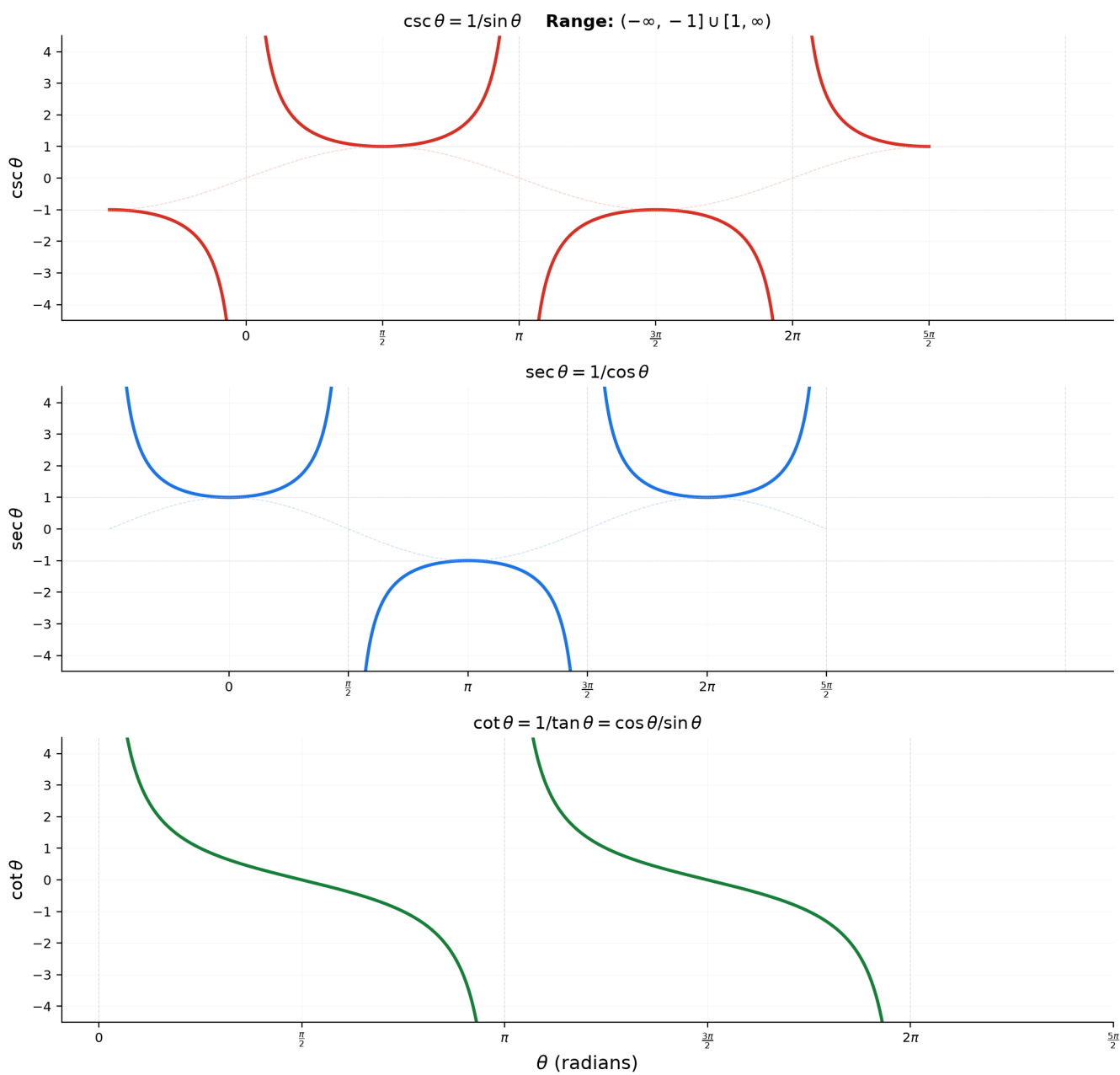
- Same shape as $\csc \theta$ but shifted by $\frac{\pi}{2}$.
- Asymptotes at $\theta = \frac{\pi}{2} + n\pi$ (same places as $\tan \theta$).
- **Period** = 2π . **Range**: $(-\infty, -1] \cup [1, \infty)$.

$\cot \theta$ **graph** (reciprocal of tangent):

- $\cot \theta = \frac{\cos \theta}{\sin \theta}$ = slope of the ray measured from the y -axis.
- Starts at $+\infty$ at 0, crosses 0 at $\frac{\pi}{2}$, goes to $-\infty$ at π .
- **Period** = π . Asymptotes at $\theta = n\pi$.
- **Range**: all real numbers.

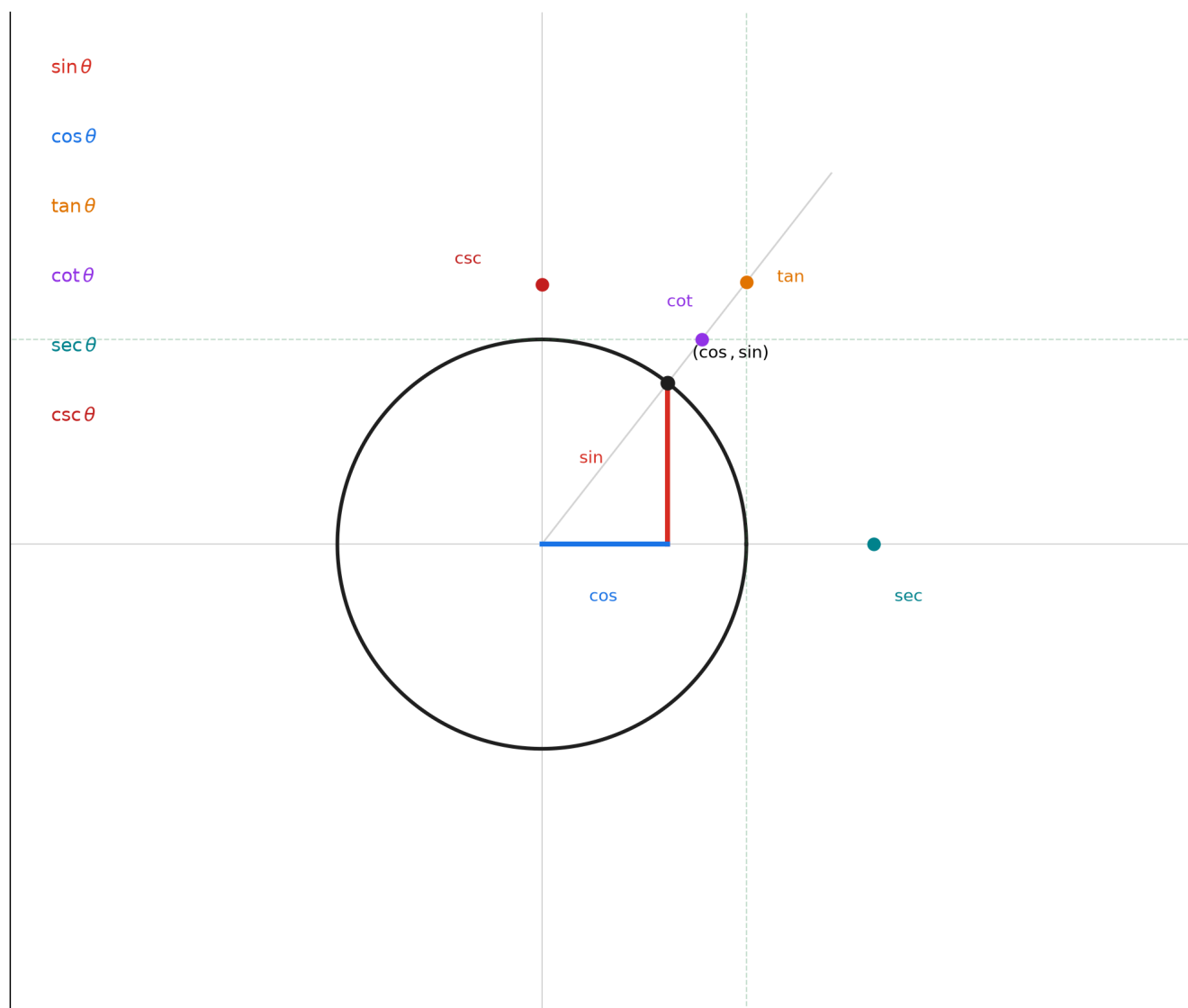
Key reciprocal relationships:

- $\csc \theta \cdot \sin \theta = 1$. $\sec \theta \cdot \cos \theta = 1$. $\cot \theta \cdot \tan \theta = 1$.
- Pythagorean variants: $1 + \tan^2 \theta = \sec^2 \theta$. $1 + \cot^2 \theta = \csc^2 \theta$.



Geometric insight on the unit circle: Draw the tangent line at $(0, 1)$ (horizontal). The ray extended hits it at $(\cot \theta, 1)$. Draw the tangent line at $(1, 0)$ (vertical). The ray hits it at $(1, \tan \theta)$. Draw the tangent at the point itself — the x -intercept of that tangent line relates to $\csc \theta$ and $\sec \theta$. All six functions come from the same diagram.

All Six Trig Functions from One Diagram



Method — Sketching $\csc$, $\sec$, $\cot$ from their reciprocals in 3 steps:

(1) **Lightly sketch the reciprocal function first.** For $\csc \theta$, sketch $\sin \theta$. For $\sec \theta$, sketch $\cos \theta$. For $\cot \theta$, sketch $\tan \theta$.

(2) **Draw dashed vertical lines where the reciprocal crosses zero.** These are the asymptotes. $\csc$: at $\theta = n\pi$ (where $\sin = 0$). $\sec$: at $\theta = \frac{\pi}{2} + n\pi$ (where $\cos = 0$). $\cot$: at $\theta = n\pi$ (where $\sin = 0$).

(3) **Flip the values: where the reciprocal is at its peak (± 1), the function touches ± 1 .** Where the reciprocal is near 0, the function shoots toward $\pm\infty$. Draw U-shaped branches between asymptotes. For $\csc$ and $\sec$, the branches always stay outside the strip $(-1, 1)$. For

$\cot$, the curve falls from $+\infty$ to $-\infty$ crossing the θ -axis once per period.

Walkthrough — sketch $\csc \theta$ on $[0, 2\pi]$: Light-draw $\sin \theta$ (starts 0, peaks at $\pi/2$, back to 0 at π , dips to -1 at $3\pi/2$, back to 0 at 2π). Asymptotes at $0, \pi, 2\pi$. At $\pi/2$ ($\sin = 1$): $\csc = 1$. At $3\pi/2$ ($\sin = -1$): $\csc = -1$. Branches: $(0, \pi)$ opens upward from $+\infty$ to 1 to $+\infty$. $(\pi, 2\pi)$ opens downward from $-\infty$ to -1 to $-\infty$.

Up to here: $\sin$ and $\cos$ are the y and x of a point on the unit circle. $\tan$ is the slope. $\csc$, $\sec$, $\cot$ are their reciprocals. Each has a characteristic graph with period and asymptotes (except $\sin$ and $\cos$, which have no asymptotes).

Example 9: Transforming the Graphs — Stretch, Shift, Compress

Given $y = A \sin(B\theta - C) + D$ (and same form for $\cos$):

- $|A|$ = **amplitude**: the wave reaches $|A|$ above and below the midline. If $A < 0$, the wave flips vertically.
- $|B|$ = **frequency factor**: period = $\frac{2\pi}{|B|}$. $B = 2$ squeezes the wave (period = π). $B = \frac{1}{2}$ stretches it (period = 4π).
- $\frac{C}{B}$ = **phase shift**: horizontal shift. Positive = shift right.
- D = **vertical shift**: the midline moves to $y = D$.

Example — build $y = 2 \sin(3\theta - \frac{\pi}{2}) + 1$ step by step:

(1) Start with $y = \sin \theta$. Period 2π , amplitude 1, midline $y = 0$.

(2) $y = 2 \sin \theta$. Amplitude becomes 2. Wave stretches vertically from -2 to 2.

(3) $y = 2 \sin(3\theta)$. $B = 3$. Period = $\frac{2\pi}{3}$. Wave compresses horizontally — three full waves in $[0, 2\pi]$.

(4) $y = 2 \sin(3\theta - \frac{\pi}{2})$. Phase shift = $\frac{\pi/2}{3} = \frac{\pi}{6}$ to the right.

(5) $y = 2 \sin(3\theta - \frac{\pi}{2}) + 1$. Shift everything up by 1. Midline is now $y = 1$. Range: $[-1, 3]$.

For tan: $y = A \tan(B\theta - C) + D$. Period = $\frac{\pi}{|B|}$. Asymptotes shift with the phase shift. No "amplitude" (wave is unbounded), but $|A|$ controls vertical stretch.

Example — build $y = \frac{1}{2} \tan(2\theta + \frac{\pi}{4}) - 1$ step by step:

(1) Start with $y = \tan \theta$. Period π , asymptotes at $\frac{\pi}{2} + n\pi$, crosses 0 at $n\pi$.

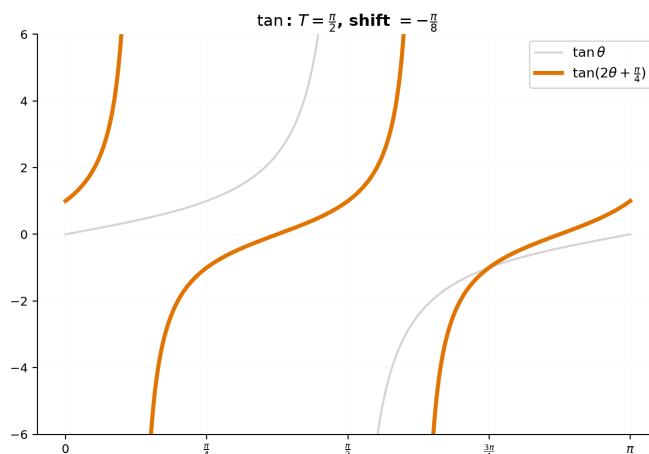
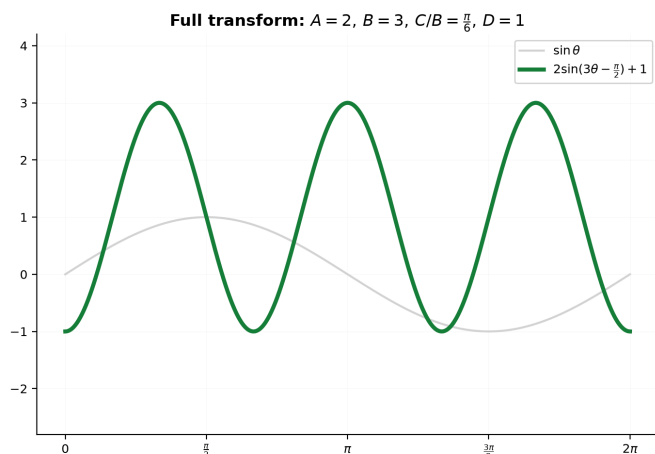
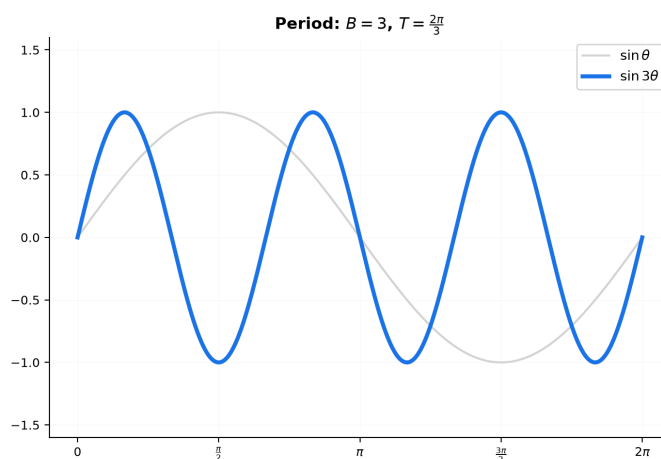
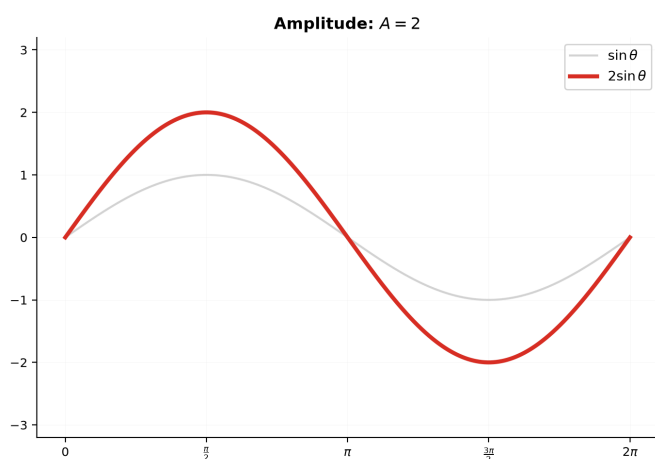
(2) $y = \tan(2\theta)$. $B = 2$. Period = $\frac{\pi}{2}$. Asymptotes at $\frac{\pi}{4} + n\frac{\pi}{2}$. Compresses horizontally.

(3) $y = \tan(2\theta + \frac{\pi}{4})$. Rewrite: $y = \tan(2(\theta + \frac{\pi}{8}))$. Phase shift = $-\frac{\pi}{8}$ (left). All asymptotes and zero-crossings shift left by $\frac{\pi}{8}$.

(4) $y = \frac{1}{2} \tan(2\theta + \frac{\pi}{4})$. Vertical compression by factor $\frac{1}{2}$. The S-curve is half as tall at every point.

(5) $y = \frac{1}{2} \tan(2\theta + \frac{\pi}{4}) - 1$. Shift down by 1. Midline (the "center" of the S) moves from $y = 0$ to $y = -1$. Zero-crossings become crossings at $y = -1$.

Transformations: $A \cdot \text{trig}(B\theta - C) + D$, Step by Step



Geometric insight: A stretches the circle's radius vertically. B changes how fast the point spins around the circle. C changes where you start. D raises or lowers the center of the circle.

Part C: Inverse Trigonometric Functions — Flipping the Question

Example 10: $\arcsin x$ — "What Angle Gives This Sine?"

$\sin \theta = x \rightarrow \theta = \arcsin x$ (also written $\sin^{-1} x$, but this is NOT $\frac{1}{\sin x}$).

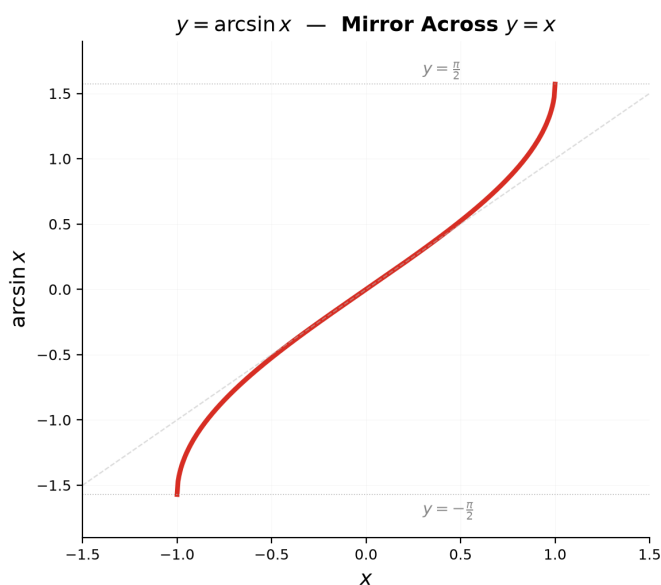
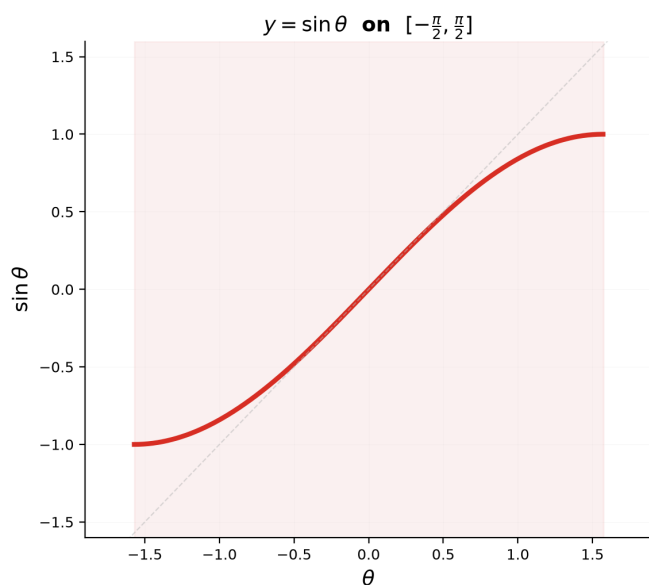
The problem: $\sin \theta$ is not one-to-one. $\sin 0 = 0$, $\sin \pi = 0$, $\sin 2\pi = 0$. To define an inverse, we must **restrict the domain**.

For $\arcsin$, restrict $\sin$ to $[-\frac{\pi}{2}, \frac{\pi}{2}]$. On this interval, sine rises strictly from -1 to 1 — every value appears exactly once.

Domain of $\arcsin$: $[-1, 1]$. **Range of $\arcsin$:** $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

$$\arcsin 0 = 0. \arcsin \frac{1}{2} = \frac{\pi}{6}. \arcsin 1 = \frac{\pi}{2}. \arcsin(-1) = -\frac{\pi}{2}.$$
$$\arcsin \frac{\sqrt{2}}{2} = \frac{\pi}{4}. \arcsin(-\frac{\sqrt{3}}{2}) = -\frac{\pi}{3}.$$

Graph: Take the graph of $y = \sin \theta$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, flip it across the line $y = x$. The result is a curve that rises from $(-1, -\frac{\pi}{2})$ to $(1, \frac{\pi}{2})$, steepest at the ends, flattest at the origin.



Geometric insight: $\arcsin x$ answers "at what angle (between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$) is the y -coordinate on the unit circle equal to x ?" The restriction to $[-\frac{\pi}{2}, \frac{\pi}{2}]$ picks the *right half* of the unit circle — where every horizontal line hits the circle exactly once.

Example 11: $\arccos x$ — "What Angle Gives This Cosine?"

For $\arccos$, restrict $\cos$ to $[0, \pi]$. On this interval, cosine falls strictly from 1 to -1 .

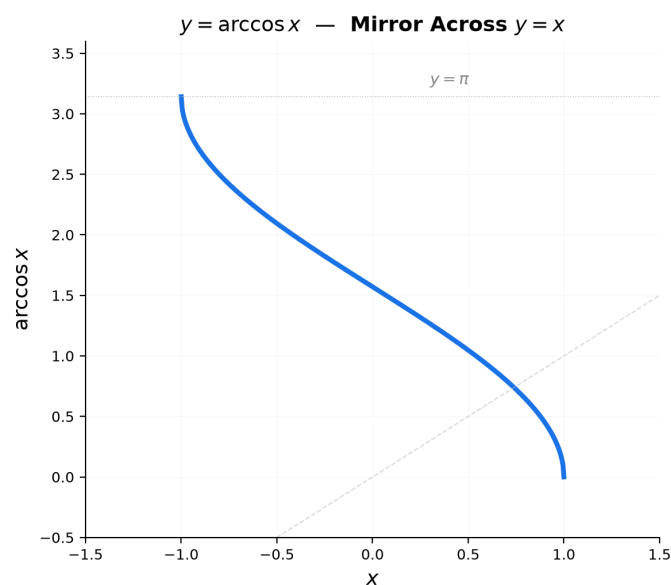
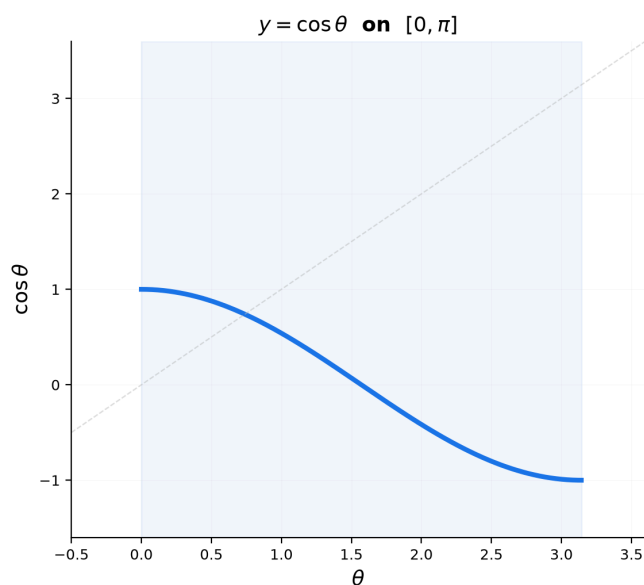
Domain of $\arccos$: $[-1, 1]$. **Range of $\arccos$:** $[0, \pi]$.

$$\arccos 1 = 0. \arccos \frac{1}{2} = \frac{\pi}{3}. \arccos 0 = \frac{\pi}{2}. \arccos(-1) = \pi.$$

$$\arccos \frac{\sqrt{2}}{2} = \frac{\pi}{4}. \arccos(-\frac{\sqrt{3}}{2}) = \frac{5\pi}{6}.$$

Graph: Flip $y = \cos \theta$ on $[0, \pi]$ across $y = x$. Falls from $(-1, \pi)$ to $(1, 0)$. Steepest at the ends, flattest at the middle.

Key relationship: $\arcsin x + \arccos x = \frac{\pi}{2}$ for all $x \in [-1, 1]$. The two angles are complementary.



Geometric insight: $\arccos x$ answers "at what angle (between 0 and π) is the x -coordinate on the unit circle equal to x ?" The restriction to $[0, \pi]$ picks the *top half* of the unit circle.

Example 12: $\arctan x$ — "What Angle Gives This Slope?"

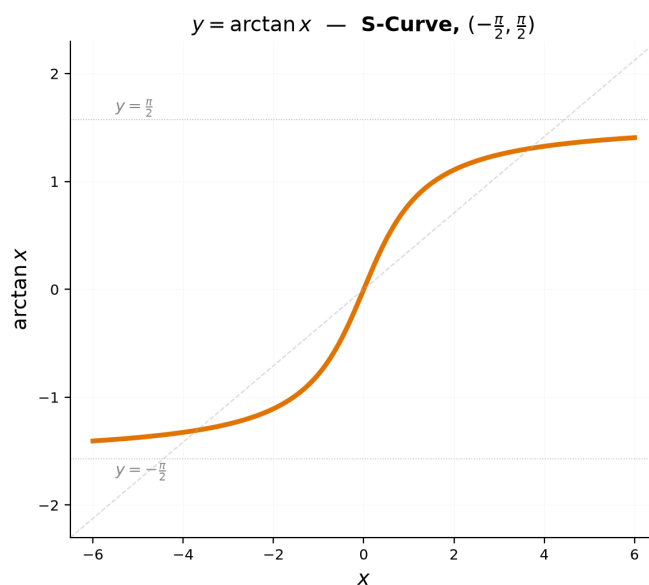
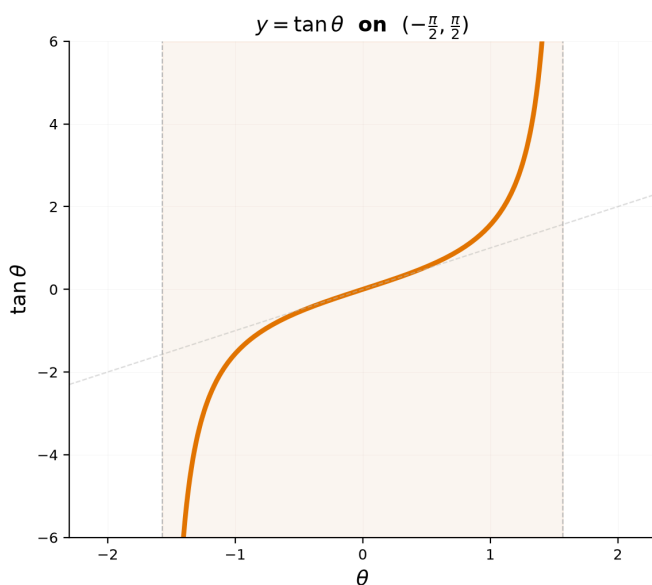
For $\arctan$, restrict $\tan$ to $(-\frac{\pi}{2}, \frac{\pi}{2})$. On this interval, $\tan$ rises from $-\infty$ to $+\infty$, hitting every real number exactly once.

Domain of $\arctan$: all real numbers. **Range of $\arctan$:** $(-\frac{\pi}{2}, \frac{\pi}{2})$.

$$\arctan 0 = 0. \arctan 1 = \frac{\pi}{4}. \arctan \sqrt{3} = \frac{\pi}{3}. \arctan(-1) = -\frac{\pi}{4}.$$

As $x \rightarrow \infty$, $\arctan x \rightarrow \frac{\pi}{2}$ (but never reaches it). As $x \rightarrow -\infty$, $\arctan x \rightarrow -\frac{\pi}{2}$.

Graph: S-shaped curve with horizontal asymptotes at $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$. Passes through $(0, 0)$. Steepest at the origin, flattening toward the asymptotes.



Geometric insight: $\arctan x$ answers "at what angle (between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$) does the ray have slope x ?" The restriction to $(-\frac{\pi}{2}, \frac{\pi}{2})$ picks the *right half* of the unit circle, excluding the vertical position where slope is undefined.

Example 13: Composing Trig and Inverse Trig — Watch the Domains!

$\arcsin(\sin \theta)$: Only equals θ if $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

$\arcsin(\sin \frac{\pi}{6}) = \arcsin(\frac{1}{2}) = \frac{\pi}{6}$. Fine — $\frac{\pi}{6}$ is in range.

$\arcsin(\sin \frac{5\pi}{6})$: $\sin \frac{5\pi}{6} = \frac{1}{2}$. $\arcsin(\frac{1}{2}) = \frac{\pi}{6}$. **Not $\frac{5\pi}{6}$!**

Why: $\frac{5\pi}{6}$ is outside $\arcsin$'s range. The answer is the angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ with the same sine — which is $\frac{\pi}{6}$.

$\arcsin(\sin \frac{7\pi}{6})$: $\sin \frac{7\pi}{6} = -\frac{1}{2}$. $\arcsin(-\frac{1}{2}) = -\frac{\pi}{6}$. (Not $\frac{7\pi}{6}$.)

$\sin(\arcsin x)$: Always equals x , for all $x \in [-1, 1]$. The outer $\sin$ undoes the inner $\arcsin$.

$\sin(\arcsin \frac{3}{5}) = \frac{3}{5}$. $\sin(\arcsin(-0.7)) = -0.7$.

Method — Composing inverse-then-trig (right-triangle method), 3 steps:

Given something like $\cos(\arcsin x)$ or $\tan(\arccos x)$, you are answering: "Take the angle

whose sine is x . What is its cosine?"

(1) **Draw a right triangle.** Label the angle as θ . Write the given ratio on two sides. For $\arcsin x$: opposite = x , hypotenuse = 1. For $\arccos x$: adjacent = x , hypotenuse = 1. For $\arctan x$: opposite = x , adjacent = 1.

(2) **Find the missing side with Pythagoras.** $a^2 + b^2 = c^2$. For $\arcsin x$: adjacent = $\sqrt{1 - x^2}$ (positive, since $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ guarantees $\cos \geq 0$).

(3) **Read the target trig function from the triangle.** $\cos \theta = \text{adjacent/hypotenuse}$. $\tan \theta = \text{opposite/adjacent}$. $\sin \theta = \text{opposite/hypotenuse}$.

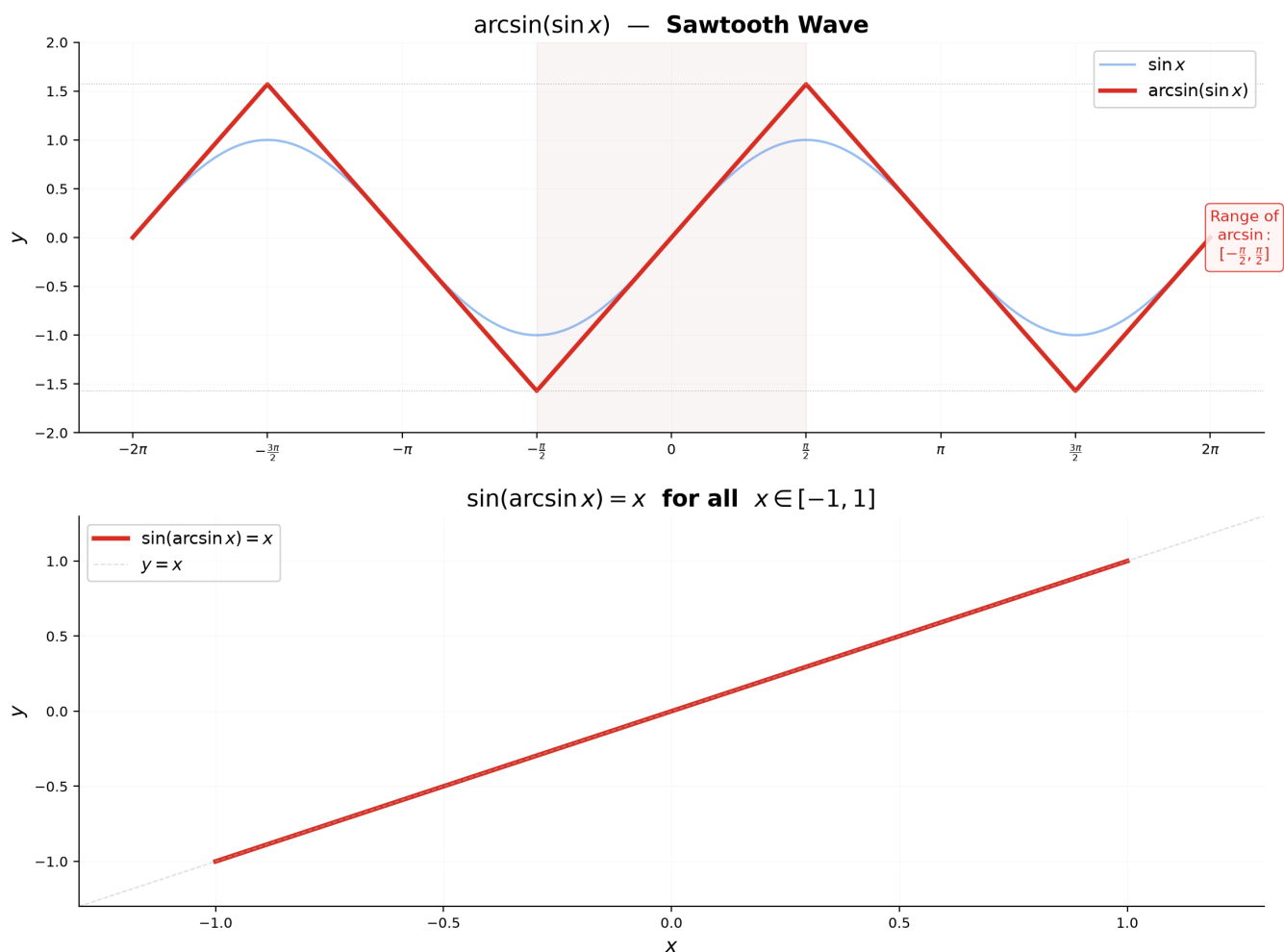
Walkthrough 1 — $\cos(\arcsin \frac{3}{5})$: (1) Triangle: opposite = 3, hypotenuse = 5. (2) Adjacent = $\sqrt{25 - 9} = 4$. (3) $\cos \theta = \frac{4}{5}$.

Walkthrough 2 — $\tan(\arccos(-\frac{5}{13}))$: (1) Triangle: adjacent = -5 (QII angle), hypotenuse = 13. (2) Opposite = $\sqrt{169 - 25} = 12$. Since $\theta \in [0, \pi]$ and $\cos \theta = -\frac{5}{13}$, θ is in QII $\rightarrow \sin \theta > 0$, so opposite = $+12$. (3) $\tan \theta = \frac{12}{-5} = -\frac{12}{5}$.

Walkthrough 3 — $\sin(\arctan \frac{3}{4})$: (1) Triangle: opposite = 3, adjacent = 4. (2) Hypotenuse = $\sqrt{9 + 16} = 5$. (3) $\sin \theta = \frac{3}{5}$.

Walkthrough 4 — $\cos(\arcsin x)$ as a formula: (1) Triangle: opposite = x , hypotenuse = 1. (2) Adjacent = $\sqrt{1 - x^2}$. (3) $\cos \theta = \frac{\sqrt{1-x^2}}{1} = \sqrt{1 - x^2}$.

Walkthrough 5 — $\tan(\arcsin x)$ as a formula: (1) Triangle: opposite = x , hypotenuse = 1. (2) Adjacent = $\sqrt{1 - x^2}$. (3) $\tan \theta = \frac{x}{\sqrt{1-x^2}}$.



Geometric insight: $\arcsin(\sin \theta)$ produces a "sawtooth" wave — it bounces θ back into $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Think of folding the number line like an accordion into that interval. $\sin(\arcsin x)$ is just the identity on $[-1, 1]$. $\cos(\arcsin x)$ draws a right triangle and uses Pythagoras.

Up to here: $\arcsin$, $\arccos$, $\arctan$ are inverses with restricted domains. Their graphs are mirrors of the original across $y = x$. Composition requires checking whether the angle is in the restricted range; if not, find the equivalent angle that is.

Method — Sketching arcsin, arccos, arctan in 3 steps:

(1) **Draw the original function on its restricted (principal) branch.** $\sin$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$: rises from -1 to 1 . $\cos$ on $[0, \pi]$: falls from 1 to -1 . $\tan$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$: rises from $-\infty$ to $+\infty$, crossing 0 .

(2) **Draw the line $y = x$ (the mirror).** For each key point (a, b) on the original, plot the reflected point (b, a) .

sin: $(-\frac{\pi}{2}, -1) \rightarrow (-1, -\frac{\pi}{2}), (0, 0) \rightarrow (0, 0), (\frac{\pi}{2}, 1) \rightarrow (1, \frac{\pi}{2})$.

cos: $(0, 1) \rightarrow (1, 0), (\frac{\pi}{2}, 0) \rightarrow (0, \frac{\pi}{2}), (\pi, -1) \rightarrow (-1, \pi)$.

tan: horizontal asymptotes of tan become vertical bounds for arctan: $y \rightarrow \pm \frac{\pi}{2}$.

(3) **Connect the reflected points with the characteristic shape.** arcsin: steep at ends, flat at center. arccos: steep at ends, flat at center (falling). arctan: S-curve, steepest at origin, flattening toward $\pm \frac{\pi}{2}$.

Walkthrough — sketch arccos x : Start with $\cos \theta$ on $[0, \pi]$. Key points: $(0, 1), (\frac{\pi}{3}, \frac{1}{2}), (\frac{\pi}{2}, 0), (\frac{2\pi}{3}, -\frac{1}{2}), (\pi, -1)$. Reflect across $y = x$: $(1, 0), (\frac{1}{2}, \frac{\pi}{3}), (0, \frac{\pi}{2}), (-\frac{1}{2}, \frac{2\pi}{3}), (-1, \pi)$. Connect smoothly falling from left to right.

Decision Tree — Evaluating Trig at Any Angle

You need $\sin \theta, \cos \theta, \tan \theta$ for an angle θ :

- └ (1) Is θ one of the 5 special angles in QI? ($0, \pi/6, \pi/4, \pi/3, \pi/2$)
 - └ Read exact value from the table. Done.
 - └ (2) Find the reference angle α — the acute angle to the x-axis.
 - └ If $\theta > 2\pi$, subtract 2π repeatedly until $\theta \in [0, 2\pi)$.
 - └ If $\theta < 0$, add 2π repeatedly until $\theta \in [0, 2\pi)$.
 - └ QI ($0 < \theta < \pi/2$): $\alpha = \theta$
 - └ QII ($\pi/2 < \theta < \pi$): $\alpha = \pi - \theta$
 - └ QIII ($\pi < \theta < 3\pi/2$): $\alpha = \theta - \pi$
 - └ QIV ($3\pi/2 < \theta < 2\pi$): $\alpha = 2\pi - \theta$
 - └ (3) $\sin \theta = \pm \sin \alpha, \cos \theta = \pm \cos \alpha$. Sign from ASTC quadrant.
 - └ (4) $\tan \theta = \sin \theta / \cos \theta$.
-

Decision Tree — Graphing a Trig Function

Given $y = A \cdot \text{trig}(B(\theta - C)) + D$:

- |— (1) Identify the parent function (sin, cos, tan, csc, sec, cot).
 - | — Draw its basic shape with period and key points.
- |— (2) Apply vertical stretch $|A|$ and flip if $A < 0$.
 - | — sin/cos: amplitude = $|A|$. tan/cot/sec/csc: vertical stretch only.
- |— (3) Apply horizontal stretch/compression — period = (base period)/ $|B|$.
 - | — sin/cos/csc/sec: base = 2π . tan/cot: base = π .
- |— (4) Apply horizontal shift C (right if positive).
- |— (5) Apply vertical shift D (up if positive).
- |— (6) Draw asymptotes for tan/cot/sec/csc:
 - | — tan: $\theta = C + (\pi/2|B|) + n \cdot (\pi/|B|)$
 - | — cot: $\theta = C + n \cdot (\pi/|B|)$
 - | — sec: $\theta = C + (\pi/2|B|) + n \cdot (2\pi/|B|)$
 - | — csc: $\theta = C + n \cdot (2\pi/|B|)$
- |— (7) Plot 5 key points per period and connect smoothly.

Common Mistakes

Mistake 1: Confusing $\sin^{-1} x$ with $(\sin x)^{-1} = \csc x$

Wrong path: Seeing $\sin^{-1} x$ and thinking it means $\frac{1}{\sin x}$.

Why wrong: $\sin^{-1} x$ is the *inverse function* (arcsine), not the reciprocal. This is an unfortunate notation collision. $\sin^2 x$ means $(\sin x)^2$, but $\sin^{-1} x$ does NOT mean $(\sin x)^{-1}$.

Right path: $\sin^{-1} x = \arcsin x$ (the angle whose sine is x). $\frac{1}{\sin x} = \csc x$ (the reciprocal).
When in doubt, use $\arcsin$ notation to avoid confusion.

Mistake 2: Forgetting that arcsin output is restricted to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Wrong path: " $\arcsin(\sin \frac{3\pi}{4}) = \frac{3\pi}{4}$."

Why wrong: $\frac{3\pi}{4}$ is not in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. arcsin can only output angles in this range.

Right path: $\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}$. $\arcsin \frac{\sqrt{2}}{2} = \frac{\pi}{4}$. The correct answer is $\frac{\pi}{4}$.

Mistake 3: Using degrees in calculus formulas

Wrong path: " $\frac{d}{dx} \sin x^\circ = \cos x^\circ$."

Why wrong: The derivative formula $\frac{d}{dx} \sin x = \cos x$ is only true when x is in radians. With degrees, you get an extra factor of $\frac{\pi}{180^\circ}$.

Right path: Always convert to radians before doing calculus. $\frac{d}{dx} \sin x = \cos x$ (radians only).

Mistake 4: Mixing up period of tan with sin/cos

Wrong path: " $\tan \theta$ has period 2π ."

Why wrong: $\tan \theta = \tan(\theta + \pi)$, not $\tan(\theta + 2\pi)$. The slope repeats after a half-turn.

Right path: $\tan \theta$ and $\cot \theta$ have period π . $\sin \theta$, $\cos \theta$, $\sec \theta$, $\csc \theta$ have period 2π .

Mistake 5: Drawing csc θ and sec θ graphs with values between -1 and 1

Wrong path: Drawing csc θ that dips to 0 between asymptotes.

Why wrong: $\csc \theta = \frac{1}{\sin \theta}$. Since $|\sin \theta| \leq 1$, $|\csc \theta| \geq 1$. It can never be between -1 and 1 .

Right path: $\csc \theta$ and $\sec \theta$ live entirely in $(-\infty, -1] \cup [1, \infty)$. Their graphs never cross the strip between $y = -1$ and $y = 1$.

What We Just Did

- (1) Radians – defined by arc length: $s = r\theta$. $360^\circ = 2\pi$ rad. Convert: multiply by $\pi/180^\circ$ or $180^\circ/\pi$. Method: (i) pick direction, (ii) multiply + reduce fraction, (iii) wrap in $[0, 2\pi)$ if needed. Radians are the natural unit; calculus only works in radians.
 - (2) Unit circle – $\cos \theta = x$ -coordinate, $\sin \theta = y$ -coordinate of a point on $x^2+y^2=1$. Special angles ($\pi/6, \pi/4, \pi/3, \pi/2$) have exact radical coordinates. Method for any angle: (i) normalize to $[0, 2\pi)$, (ii) find reference angle α , (iii) read $\sin \alpha / \cos \alpha$ from table, apply ASTC sign. $\tan = \sin / \cos$.
 - (3) Six functions – $\sin$ and $\cos$ are waves of period 2π , amplitude 1. $\tan = \sin / \cos$, period π , vertical asymptotes where $\cos=0$. $\csc = 1/\sin$, $\sec = 1/\cos$, $\cot = 1/\tan$ – each has asymptotes where the denominator is 0. Method for $\csc/\sec/\cot$ graphs: (i) lightly sketch the reciprocal, (ii) draw asymptotes where reciprocal=0, (iii) flip values (large $\leftrightarrow$ small). Branches never enter $(-1,1)$.
 - (4) Graph transformations – A = vertical stretch, B = frequency factor (period = base/ $|B|$), C/B = phase shift, D = vertical shift. Method (5 steps): parent $\rightarrow A \rightarrow B \rightarrow C \rightarrow D$. Applied to both $\sin/\cos$ and $\tan$ with their respective base periods.
 - (5) Inverse functions – $\arcsin: [-1,1] \rightarrow [-\pi/2, \pi/2]$. $\arccos: [-1,1] \rightarrow [0, \pi]$. $\arctan: \mathbb{R} \rightarrow (-\pi/2, \pi/2)$. Method for sketching: (i) draw original on restricted branch (ii) reflect key points across $y=x$, (iii) connect with characteristic shape.
 - (6) Composition – $\arcsin(\sin \theta) = \theta$ only if $\theta \in [-\pi/2, \pi/2]$; otherwise fold it in. $\sin(\arcsin x) = x$ always (for $x \in [-1,1]$). Method for $\cos(\arcsin x)$ etc.: (i) draw right triangle, label given ratio on two sides (ii) find missing side with Pythagoras, (iii) read target function from triangle sides.
-

Practice 1

Convert: (a) 150° to radians. (b) $\frac{7\pi}{6}$ to degrees. (c) -45° to radians.

→ Reference: **Example 2**

Solutions: [Solutions](#)

Practice 2

Find the exact values without a calculator:

(a) $\sin \frac{5\pi}{3}$ (b) $\cos \frac{3\pi}{4}$ (c) $\tan \frac{7\pi}{6}$ (d) $\sec \frac{5\pi}{4}$

→ Reference: **Example 4, 5, 7, 8**

Solutions: [Solutions](#)

Practice 3

Sketch the graph of $y = -3 \cos(2\theta + \frac{\pi}{3}) - 1$. Label: amplitude, period, phase shift, midline, and the five key points of one period.

→ Reference: **Example 9**

Solutions: [Solutions](#)

Practice 4

Evaluate: (a) $\arcsin(-\frac{\sqrt{3}}{2})$ (b) $\arccos(-\frac{\sqrt{2}}{2})$ (c) $\arctan(-\sqrt{3})$

→ Reference: **Example 10, 11, 12**

Solutions: [Solutions](#)

Practice 5: Composition

Find the exact value of $\sin(\arccos \frac{5}{13} + \arcsin \frac{4}{5})$. Draw right triangles for each inverse trig expression.

→ Reference: **Example 13**

Solutions: [Solutions](#)

Practice 6

Simplify: (a) $\cos(\arctan x)$ as an algebraic expression in x . (b) $\tan(\arcsin \frac{x}{\sqrt{x^2+1}})$.

→ Reference: **Example 13**

Solutions: [Solutions](#)

Practice 7

Identify the domain restrictions for: (a) $\arcsin(2x - 1)$ (b) $\arccos(\frac{x}{x+1})$. Give your answer in interval notation.

→ Reference: **Example 10, 11**

Solutions: [Solutions](#)

Practice 8: Composition — Build the Graph

Draw the graph of $y = \arcsin(\sin x)$ for $x \in [-2\pi, 2\pi]$. Explain the sawtooth pattern. Then draw $y = \sin(\arcsin x)$ on its natural domain.

→ Reference: **Example 13**

Solutions: [Solutions](#)

Practice 9: Real Battle — From Circle to Graph to Inverse

A point P moves counterclockwise on the unit circle, starting at $(1, 0)$ at $t = 0$, completing one revolution every 4π seconds.

- (a) Write the y -coordinate of P as a function of time: $y(t) = \sin(Bt)$. Find B .
- (b) Write the x -coordinate as $x(t) = \cos(Bt)$.
- (c) At what times $t \in [0, 4\pi]$ does the point cross the line $y = \frac{1}{2}$? Give exact answers.
- (d) What is the slope of the ray at $t = \frac{\pi}{3}$?
- (e) At what angle (in radians) does the ray have slope 2? Give your answer using $\arctan$.

→ Reference: **Example 3, 4, 6, 7, 10**

Solutions: [Solutions](#)

Practice 10: Real Battle — Graph All Six

On the same set of axes $[0, 2\pi]$, sketch $\sin \theta$, $\cos \theta$, and $\tan \theta$. Use three different colors. Mark all asymptotes for $\tan \theta$. Then on a separate set of axes, sketch $\csc \theta$, $\sec \theta$, and $\cot \theta$.

→ Reference: **Example 6, 7, 8**

Solutions: [Solutions](#)

Basic Algebra Drill — Radians and Values (10 Problems)

Pure fluency. Build instant recall of special angles and values.

D1. Convert 210° to radians. Write as a fraction of π .

D2. Convert $\frac{11\pi}{6}$ to degrees.

D3. What is the reference angle for $\frac{5\pi}{4}$?

D4. $\sin \frac{2\pi}{3} = ?$ (exact value)

D5. $\cos \frac{5\pi}{6} = ?$ (exact value)

D6. $\tan \frac{4\pi}{3} = ?$ (exact value)

D7. $\sec \frac{\pi}{3} = ?$ (exact value)

D8. $\csc \frac{7\pi}{6} = ?$ (exact value)

D9. $\cot \frac{3\pi}{4} = ?$ (exact value)

D10. $\arcsin(\sin \frac{7\pi}{4}) = ?$

Solutions: [Solutions](#)

Advanced Algebra Drill — Graphs, Inverses, and Composition (10 Problems)

Multi-step. Each problem chains 2–3 skills from 11A. Covers graph transformations, inverse

evaluation, and composition.

- A1.** Graph $y = 2 \csc(\theta - \frac{\pi}{4})$ on $[0, 2\pi]$. Label all asymptotes and the coordinates of all local minima and maxima.
- A2.** Graph $y = -2 \sin(\frac{1}{2}\theta + \frac{\pi}{3})$ on $[-2\pi, 4\pi]$. Identify amplitude, period, phase shift, and midline. Mark all x -intercepts.
- A3.** Graph $y = 3 \tan(\frac{\theta}{2} - \frac{\pi}{6})$ on $[-\pi, 3\pi]$. Draw all asymptotes. Mark where the curve crosses the midline.
- A4.** Evaluate $\sin(\arccos \frac{12}{13}) + \cos(\arcsin \frac{3}{5})$ without a calculator. (Hint: draw two right triangles.)
- A5.** Simplify $\sec(\arctan x)$ as an algebraic expression in x . Use the right-triangle method.
- A6.** Find the exact value of $\arcsin(\cos \frac{5\pi}{6})$. (Hint: convert cos to sin first.)
- A7.** Given $\arcsin x = \arccos 2x$ for $x > 0$, find x exactly. (Hint: take sine of both sides.)
- A8.** A point P on the unit circle has coordinates $(\frac{5}{13}, -\frac{12}{13})$. (a) What is $\sin \theta$ and $\cos \theta$? (b) In which quadrant is θ ? (c) Find $\tan \theta$, $\csc \theta$, $\sec \theta$, $\cot \theta$. (d) What is the reference angle? (e) Write θ as an inverse trig expression.
- A9.** The function $f(\theta) = 4 \sin(3\theta - \frac{\pi}{2}) + 2$ models the height of a Ferris wheel car (in meters, θ in minutes). (a) What is the maximum height? (b) At what $\theta \in [0, 2\pi]$ does it first reach the maximum? (c) What is the period, and what does it mean physically? (d) Sketch one full period with labels.
- A10.** Find all $x \in [-1, 1]$ such that $\arcsin x = \arccos x$. Then solve $\arctan x = \arccos x$ for $x > 0$. Give exact answers.

Solutions: [Solutions](#)

Today's Procedure

- Step 1: Angle conversion – degrees→rad: multiply by $\pi/180^\circ$, reduce fraction, wrap into $[0, 2\pi)$
rad→degrees: multiply by $180^\circ/\pi$. Radians = arc length / radius = natural angle unit
- Step 2: Any trig value – normalize angle to $[0, 2\pi)$. Find reference angle α by quadrant.
Read $\sin \alpha$, $\cos \alpha$ from the $\sqrt{n}/2$ table. Apply ASTC sign. $\tan = \sin/\cos$.
 $\csc = 1/\sin$. $\sec = 1/\cos$. $\cot = 1/\tan$.
- Step 3: Graph $\sin/\cos/\tan$ – identify parent. Apply A (vertical), B (period = $2\pi/|B|$), C (phase shift = C/B), D (vertical shift) in sequence. Plot 5 key points per period.
For $\tan$: base period = π , draw asymptotes at $\theta = C/B + \pi/(2|B|) + n \cdot \pi/|B|$.
- Step 4: Graph $\csc/\sec/\cot$ – lightly sketch the reciprocal ($\sin/\cos/\tan$). Draw asymptotes where reciprocal = 0. Flip values: where reciprocal = ± 1 , function = ± 1 .
Where reciprocal $\rightarrow 0$, function $\rightarrow \pm\infty$. $\csc/\sec$ never enter $(-1, 1)$.
- Step 5: Inverse trig – $\arcsin(x) \in [-\pi/2, \pi/2]$, $\arccos(x) \in [0, \pi]$, $\arctan(x) \in (-\pi/2, \pi/2)$
To sketch: restrict original to principal branch, reflect across $y=x$, connect.
To evaluate $\arcsin(\sin \theta)$: if θ outside range, find equivalent angle inside range.
- Step 6: Composition via right triangle – for $\cos(\arcsin x)$, $\tan(\arccos x)$, etc.:
(i) Draw triangle, label the given ratio on two sides.
(ii) Find the missing side with $a^2 + b^2 = c^2$.
(iii) Read the target function from the three sides.

How to Read These Symbols

Symbol	Reads as	Meaning
π	"pi"	ratio of circumference to diameter ≈ 3.14159 ; π rad = 180°
θ	"theta"	a variable representing an angle (in radians by default)

Symbol	Reads as	Meaning
$\sin \theta$	"sine theta"	y -coordinate of the point on the unit circle at angle θ
$\cos \theta$	"cosine theta"	x -coordinate of the point on the unit circle at angle θ
$\tan \theta$	"tangent theta"	$\sin \theta / \cos \theta$; slope of the ray at angle θ
$\csc \theta$	"cosecant theta"	$1 / \sin \theta$; reciprocal of sine
$\sec \theta$	"secant theta"	$1 / \cos \theta$; reciprocal of cosine
$\cot \theta$	"cotangent theta"	$1 / \tan \theta = \cos \theta / \sin \theta$; reciprocal of tangent
$\arcsin x$	"arcsine of x "	the angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ whose sine is x
$\arccos x$	"arccosine of x "	the angle in $[0, \pi]$ whose cosine is x
$\arctan x$	"arctangent of x "	the angle in $(-\frac{\pi}{2}, \frac{\pi}{2})$ whose tangent is x
ASTC	"All Students Take Calculus"	memory aid: All positive in QI, Sine in QII, Tangent in QIII, Cosine in QIV
reference angle	"reference angle"	acute angle between the terminal ray and the x -axis
period	"period"	horizontal length of one full cycle before the graph repeats
amplitude	"amplitude"	half the distance between maximum and minimum (for sin/cos only)
asymptote	"asymptote"	a vertical line the graph approaches but never crosses

Terminology

Up to now we used plain words like "height on the circle", "slope of the ray", "flip across the line $y = x$ ".

You have already learned all the methods. Now we attach the formal mathematical names.

What we called it	Mathematical term	Notation
arc-length angle	radian	rad (dimensionless; 2π rad = 360°)
circle of radius 1	unit circle	$x^2 + y^2 = 1$
x -coordinate on unit circle	cosine	$\cos \theta$
y -coordinate on unit circle	sine	$\sin \theta$
height $\div$ horizontal	tangent	$\tan \theta = \frac{\sin \theta}{\cos \theta}$
$1 \div \text{sine}$	cosecant	$\csc \theta = \frac{1}{\sin \theta}$
$1 \div \text{cosine}$	secant	$\sec \theta = \frac{1}{\cos \theta}$
$1 \div \text{tangent}$	cotangent	$\cot \theta = \frac{1}{\tan \theta}$
acute angle to x -axis	reference angle	α (alpha)
wave height	amplitude	$ A $ in $y = A \sin(B\theta - C) + D$
horizontal cycle length	period	$\frac{2\pi}{ B }$ for sin/cos; $\frac{\pi}{ B }$ for tan/cot
horizontal shift	phase shift	$\frac{C}{B}$ in $y = A \sin(B\theta - C) + D$
middle line	midline / vertical shift	D in $y = A \sin(B\theta - C) + D$
inverse of sine	arcsine / inverse sine	$\arcsin x$ or $\sin^{-1} x$
inverse of cosine	arccosine / inverse cosine	$\arccos x$ or $\cos^{-1} x$
inverse of tangent	arctangent / inverse tangent	$\arctan x$ or $\tan^{-1} x$
restricted domain for inverse	principal branch	$[-\frac{\pi}{2}, \frac{\pi}{2}]$ for arcsin; $[0, \pi]$ for arccos; $(-\frac{\pi}{2}, \frac{\pi}{2})$ for arctan

Next: [11B — Trigonometric Identities, Equations, and Beyond](#) — where you learn to split and merge angles, solve trig equations, prove identities, and discover deep connections through Euler's formula.